

AI RETAIL DECISION ENGINE · EGYPT

Product IQ

Turn your shop data into **decisions** — not dashboards

برودكت آي كيو

حول بيانات متجرك إلى قرارات — لا لوحات معلومات

The Egyptian shop owner already has the data.

What they don't have is an answer.

Disconnected

Sales in a POS export. Stock in a spreadsheet. Suppliers on WhatsApp.

Unlocalized

Generic analytics tools are built for US retail — no EGP, no Arabic, no Ramadan.

Too Slow

"Which product is quietly killing my cash flow?" takes hours to answer.

On the demo dataset: **197,400 EGP frozen in stock nobody noticed.**

One principle, everywhere

The code does the math

pandas computes every number.
Deterministic. Reproducible. Testable.

The AI does the reasoning

The LLM explains numbers it was handed.
It never invents one.

Why it matters

- A shop owner making a 200,000 EGP purchase needs the same answer every time.
- Every AI output is labelled: AI-generated or rule-based. No guessing which.

What it finds in 3 seconds

602,378

Revenue (30d) EGP

19.1%

Margin

2.22M

Inventory EGP

197,400

DEAD capital EGP

Three findings the dashboard never surfaced

DEAD STOCK Canon EOS R50 — 82,500 EGP, zero sales in 119 days

STOCKOUT RISK Redmi Note 14 is OUT OF STOCK — and it is the #3 seller

HIDDEN OPPORTUNITY Anker is #4 by revenue but #3 by PROFIT — 32% margin, invisible on every chart

THE PRODUCT

Six engines, one dataset

Analytics

KPIs & stock risk

Product DNA

8-dimension health

Simulator

What-if pricing

CEO Report

Bilingual briefs

Research

Live web integration

Board Meeting

Multi-agent debates

Unified Central Dataset

One single source of truth powering all engines deterministically

Six engines



Analytics

KPIs, dead stock,
stock risk



Product DNA

8-dimension health
fingerprint



Simulator

What-if pricing with
real elasticity



CEO Report

Executive brief,
bilingual



Research

Live web + your
numbers



Board Meeting

4 AI agents debate
a decision

Research that refuses to lie

Live web search, joined to YOUR margins — with hard guard rails.

✗ What we refuse to do

- Compare EGP to USD listings
- Treat a page fragment as a price
- Call a 1-piece pot the same as an 11-piece set
- State a market position on 1 source

✓ What we do instead

- Classify: exact match / variant / competitor
- Egyptian retailers only for pricing
- Median of validated prices + sample size
- "Insufficient data" is a valid answer

Four specialists, one decision



Market Researcher

Live web via MCP



Data Analyst

Your pandas figures



Supply Specialist

Stock & lead times



CEO

Decides, and says which source it trusts

Live output — Canon EOS R50

DECISION — CLEAR. 20% discount for 14 days; if it does not move, return to supplier.

Frees 82,500 EGP for products that are actually selling.

UNDER THE HOOD

Built to be trusted

Enterprise-grade architecture with zero compromises on integrity.

Multi-agent validation • Full RTL layout • Secure Pandas engine



UNDER THE HOOD

Built to be trusted

Enterprise-grade architecture with zero compromises on integrity.



Multi-agent validation

Independent consensus protocol ensuring maximum analytical precision.



Full RTL layout

Native Right-to-Left design providing seamless Arabic user experiences.



Secure Pandas engine

Isolated, atomic execution layer with rigorous type-checking controls.

Architecture



Frontend

10 pages · vanilla JS · full RTL · no build step



API

FastAPI · 24 endpoints · bcrypt auth · sessions



Intelligence

LangChain + Groq · FAISS RAG · CrewAI · Tavily MCP



Data

pandas engine · SQLite memory · atomic CSV validation

24 Python modules · 10 test files · Docker · GitHub Actions CI

Where most demos cheat

COMMON SHORTCUT	WHAT IT LOOKS LIKE	PRODUCTIQ SOLUTION
Fake trend arrows	margin_change hardcoded to 0.0	Computed — or null with "no baseline"
Invented forecasts	Hardcoded 45 units/month for everything	Baseline from that product's real sales
Mock data on error	Silently shows fake numbers	Demo-mode banner, impossible to miss
False precision	"7819% above market"	Rejects outliers, states sample size

Egypt-native, not translated



RTL Mirroring

Full RTL mirroring — layout flips dynamically, not just the text



EGP Throughout

EGP currency support with Arabic product names integrated directly in the data model itself



Bilingual AI Output

Every AI output generated simultaneously in both languages in a single call



Roadmap & Seasons

Ramadan +40%, Eid +30% — forecasting seasonal variations that sales history alone cannot learn

What ships next



Seasonal calendar

Ramadan · Eid · Black Friday, Hijri-aware



EGP on every action

"These 5 moves free 180,000 EGP"



Purchase Order PDF

From advice to a document you send



Arabic CSV rescue

Accept الكمية and الصنف, ask to confirm

"Why not just ChatGPT and a spreadsheet?"



Data Integration

A chatbot cannot join sales + inventory + suppliers + pricing reliably, every time.



Designed for Truth

It will happily invent a number. Ours refuses — by design.



Historical Memory

No memory of what you asked last month, or what changed since.



Bilingual & Persistent

No persistent bilingual interface a shop actually opens each morning.

Try it



docker compose up localhost:8000

Sample Egyptian electronics dataset loads automatically. No API key needed to explore.

10

Pages, all bilingual

صفحات ثنائية اللغة بالكامل

24

API endpoints

نقاط اتصال واجهة برمجة التطبيقات

10

Test files

ملفات اختبار جاهزة للتشغيل

Analytics vs. Action



Muted Analytics

Most tools only tell you **what happened** in the past.



ProductIQ

We tell you exactly **what to do** next.

< > github.com/myler71/productIQ

Most tools tell you what happened.

ProductIQ tells you what to do.

github.com/myler71/productIQ